

005 Week 4 — Helper Functions 2 (Extension · Easy)

Topic: Petting Zoo — Helper Functions · **Course:** Theme Park Creator · **Level:** Easy · **Time:** about 45 minutes

This is the code you are learning this week. It builds a whole **petting zoo**: a fence, some trees, and animals inside.

```
def post(x, z):
    blocks.fill(OAK_FENCE, pos(x, 0, z), pos(x+15, 0, z+15), FillOperation.REPLACE)
    blocks.fill(AIR, pos(x+1, 0, z+1), pos(x+14, 0, z+14), FillOperation.REPLACE)

def tree(x, z):
    blocks.place(OAK_SAPLING, pos(x+3, 0, z+3))
    blocks.place(OAK_SAPLING, pos(x+3, 0, z+7))
    blocks.place(OAK_SAPLING, pos(x+3, 0, z+11))

def animals(x, z):
    for i in range(7):
        mobs.spawn(CHICKEN, pos(x+6, 0, z+3))

def petting_zoo(x, z):
    post(x, z)
    tree(x, z)
    animals(x, z)

petting_zoo(5, 5)
```

💡 **The big idea:** `petting_zoo` is the **boss**. It does no building itself — it just **calls** three **helper functions**, and each helper does **one** job. Write each job once, and the boss puts them together.

1 · Who Does What? 🧩

Read the code above. Fill the table.

Function	Its one job (in your words)	Boss or helper?
<code>post</code>		

Function	Its one job (in your words)	Boss or helper?
tree		
animals		
petting_zoo		

💡 A **helper** is just a function that **another** function calls. There is no special keyword — the word describes its **job**, not its code.

2 · The Hollow Fence

Look at `post`. It runs **two** fills. A big solid square, then it carves the middle back to `AIR`.

fill 1 – solid fence

```
## ## ## ## ## ##
```

```
## ## ## ## ## ##
```

```
## ## ## ## ## ##
```

```
## ## ## ## ## ##
```

```
## ## ## ## ## ##
```

(x, z) to (x+15, z+15)

fill 2 – carve middle to AIR

```
## ## ## ## ## ##
```

```
## . . . . ##
```

```
## . . . . ##
```

```
## . . . . ##
```

```
## ## ## ## ## ##
```

(x+1, z+1) to (x+14, z+14)

-->

What is left after both fills? How thick is the fence wall, and why is fill 2 moved in by exactly 1?

3 · Predict — Everything Comes From x and z 🧠

Every block is placed **relative** to x and z (like $x+3$, $z+11$). Right now the boss is called with `petting_zoo(5, 5)`.

a) The trees are at $x+3$. With `petting_zoo(5, 5)`, what is the real x value of the trees?

b) You change the last line to `petting_zoo(30, 5)`. What moves — the fence, the trees, the animals, or all of them? **Why?**

💡 This is the whole point of parameters: the boss hands x , z down to **every** helper, so the **whole zoo** moves together with one number.

4 · Spot the Bug 🐛

Look closely at `animals`. It loops 7 times, but all 7 chickens land in the **same spot**.

```
def animals(x, z):  
    for i in range(7):  
        mobs.spawn(CHICKEN, pos(x+6, 0, z+3))
```

a) Why do all 7 chickens spawn on top of each other? (Hint: look at where `i` is used... or not used.)

b) Rewrite the `mobs.spawn` line so each chicken is 2 blocks further along than the last. Use `i` in the position.

5 · Extend the Zoo 💧

Add **one new helper** that digs a little pond, then let the boss call it. Fill the blank.

```
def pond(x, z):  
    blocks.fill(WATER, pos(x+8, 0, z+9), pos(x+11, 0, z+12), FillOperation.REPLACE)  
  
def petting_zoo(x, z):  
    post(x, z)  
    tree(x, z)  
    animals(x, z)  
    ____ (x, z)          # <-- call your new pond helper
```

💡 See how easy it grows? One new helper + one new line in the boss. The boss stays short and readable no matter how big the zoo gets.

In your homework world: add the pond, then change one more thing of your own (a different animal, more trees, a bigger fence). Write what you added and what happened:

6 · Show Your Work 📹

Record **one video**, one take, on your phone. Show in order:

1 · Your code. Scroll through it. Point at the boss and the four helpers.

2 · Your zoo. Point the camera. Name the fence, the trees, the animals.

Then say these out loud, filling the blanks:

Today I built ____.

The boss function is __, **and it calls** __, *and* __.

A helper function is ____.

My favourite part was __ **because** ____.

Submit ✅

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